

Yarra Vivek

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++
Frontend: React, Next.js, Redux, React Native
Backend & Cloud: Node.js, FastAPI, PostgreSQL, MongoDB, Docker, AWS, GCP, Cloudflare Workers, CI/CD
AI & ML: PyTorch, HuggingFace, LangChain, FAISS, RAG, Gemini API, Reinforcement Learning (GRPO)

ACHIEVEMENTS

- **Grand Finalist** – Meta, PyTorch, HuggingFace, and Scaler OpenEnv Hackathon: Selected among **800** teams from **31,000+** registrations; presented live in Bangalore as a **solo participant**. 2026
- **218th Place** – HackerRank Orchestrate May 2026: Ranked among **12,885** participants from **48 countries** in a 24-hour AI Agent building challenge. 2026
- Secured **Rank 16** in Codeforces Round 1081 (Div. 2) among **13,600+** participants. 2026

EXPERIENCE

Magnumwings

Software Engineering Intern

On-site

May 2026 – Present

- **Ground Control Station (GCS):** Led end-to-end development of a real-time drone control platform using Python, FastAPI, WebSockets, and WebRTC; integrated MAVLink protocol for bidirectional autonomous flight control and live telemetry ingestion.
- **Mission Dashboard:** Architected a React-based control UI with live MAVLink telemetry feeds, real-time WebRTC video streaming, and autonomous command dispatch with continuous drone state monitoring.
- **Defense Systems:** Currently building interceptor drone software with low-latency targeting and autonomous decision-making pipelines for counter-drone interception.

QSkill – SR India

AI/ML Intern

Hybrid

Nov 2025 – May 2026

- **ML Pipelines:** Built and deployed text classification pipelines using TF-IDF with Logistic Regression, KNN, and Naive Bayes; evaluated models via F1-score, precision-recall, and confusion matrix analysis to drive iterative accuracy gains.
- **Full-Stack Inference:** Developed a Node.js/Express REST API backend with a React frontend enabling real-time model inference from raw input to structured UI; eliminated manual data handling by standardizing the end-to-end prediction workflow.

Emertxe

IoT Intern

Remote

(concurrent with QSkill) Mar–Apr 2026

- **Embedded Systems:** Developed C applications on Arduino targeting microcontroller architectures; implemented sensor interfacing, GPIO control, and real-time data acquisition following an SDLC-based project structure.
- **IoT Prototyping:** Simulated and validated end-to-end circuit prototypes in PicosimLab covering hardware-software integration, signal processing pipelines, and low-level peripheral communication under NSDC-affiliated curriculum.

PROJECTS

APEX – Adaptive Pipeline for Expert Triage | Python, Groq, LLaMA 3.1/3.3, BM25, FastEmbed | [GitHub]

- Built a **7-stage** AI ticket-routing engine with a zero-shot LLaMA-3.1-8B security layer, hybrid BM25 + MiniLM retrieval, and automated escalation fallback for sparse knowledge base scenarios.
- Achieved **sub-2ms** cached query latency via similarity-based deduplication, eliminating redundant LLM calls for **40%** of tickets; added exponential-backoff retry logic and a real-time telemetry dashboard.

DataForge Arena | Python, PyTorch, HuggingFace TRL, GRPO, Docker | [GitHub]

- Designed a GRPO-based reinforcement learning environment where a 1.5B LLM agent autonomously detects and repairs adversarially corrupted tabular data via an adaptive curriculum corruptor with ground-truth reward signals.
- Achieved **+132% reward improvement** over 300 training steps with **100%** structured JSON parse success; validated correctness across a **127-test suite** (unit, integration, stress); deployed live demo on HuggingFace Spaces.

ARGUS – Ambient AI Vision Agent | Python, FastAPI, Gemini 2.0 Flash, GCP, Docker | [GitHub]

- Built a production ambient screen-intelligence agent using Gemini 2.0 Flash Vision for real-time contextual analysis; integrated 7 Google Cloud services (Cloud Run, Firestore, GCS, Secret Manager, Cloud Logging, Cloud Build, Gemini API).
- Reduced Gemini API calls by **80%** via pixel-difference filtering; enforced OWASP-aligned security (Pydantic V2, HMAC auth, rate limiting); achieved **130+ assertions** and **80%+** code coverage across 9 test modules.

VoidVault | React 19, TypeScript, Cloudflare Workers, Supabase, PostgreSQL | [Live] | [GitHub]

- Architected a full-stack serverless social platform with REST API backend on Cloudflare Workers, cursor-paginated feeds, real-time chat, CSRF-protected session auth, and per-IP rate limiting; scaled to **500+ users** and **2,000+ posts**.

EDUCATION

Vignan's Lara Institute of Technology & Science

Bachelor of Technology – Computer Science & Engineering; GPA: 7.80/10.0

Andhra Pradesh, India

2023 – 2027